Commodity Trading Games

Final Report

Commodities Markets and Models

Master Financial Engineering, IF3 – Spring 2026

Game #1: Optimal Market-Neutral Trading

Yang & Malik (2024) – Multivariate Optimization

Game #2: Beating Passive Strategies in Bullish Markets

Palazzi (2025) – Crack Spread Pairs Trading

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Commodity Sector:	Energy
Primary Anchor:	Brent Crude Oil (ICE Futures)

Submitted on DVL – March 31, 2026

GitHub: https://github.com/MatthieuHannaGerguis/Commodities_Trading

Both notebooks are self-contained and run in Google Colab without modification. This report provides the economic reasoning, mathematical framework, critical analysis, and cross-game synthesis.

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I. Introduction

In this project, we designed and backtested two quantitative trading strategies on the energy commodity sector, with Brent Crude Oil as our anchor. Each strategy adapts a published paper from cryptocurrency to energy markets, and each addresses a different investment question.

Game #1 asks: *can we generate returns that are completely independent of oil prices?* We answer by constructing a market-neutral portfolio (zero beta to Brent) using the Optimal Trading Technique from Yang & Malik (2024). Game #2 asks: *can we beat buy-and-hold on a risk-adjusted basis?* We answer by trading the Brent-Gasoline crack spread with dynamic risk management from Palazzi (2025).

Both strategies exploit cointegration, the central concept from Lecture 3. The key finding that drives our entire project is that the crack spread (Brent vs. refined products) provides a much stronger statistical signal than any equity-based pair. Only RBOB Gasoline (RB=F) is cointegrated with Brent at the 1% level ($p = 0.0034$), while no equity pair reaches even 10% significance.

We are transparent about the limitations: Game #1's optimized strategy underperforms simpler benchmarks on raw returns, and Game #2 cannot beat buy-and-hold in a bull market. We discuss why, and what it means for real-world trading.

II. Data Architecture and Exploratory Analysis

II.a) Asset Universe and Sourcing

Our dataset contains 17 energy assets (daily adjusted close, January 2019 to March 2026, 1,822 trading days), sourced from Yahoo Finance via `yfinance`. We organized it following the "Golden Rule" (Tutorial 1, Slide 3): 1 anchor (BZ=F), 3 commodity futures (CL=F, RB=F, HO=F), 3 ETFs (XLE, USO, XOP), and 10 equities from the G10 assignment (BP, SHEL, TTE, ENI, EQNR, REP, WDS, PBR, CVX, XOM). We replaced HARL and TLW with CVX and XOM due to delisting issues.

Adding commodity futures beyond the minimum requirement was a deliberate choice. The crack spread (crude vs. products) is one of the most traded relationships in energy markets: refiners use it daily to hedge their margins, and trading desks monitor it as a core signal. In our dataset, this relationship turns out to be the only statistically significant cointegration, which validates the decision.

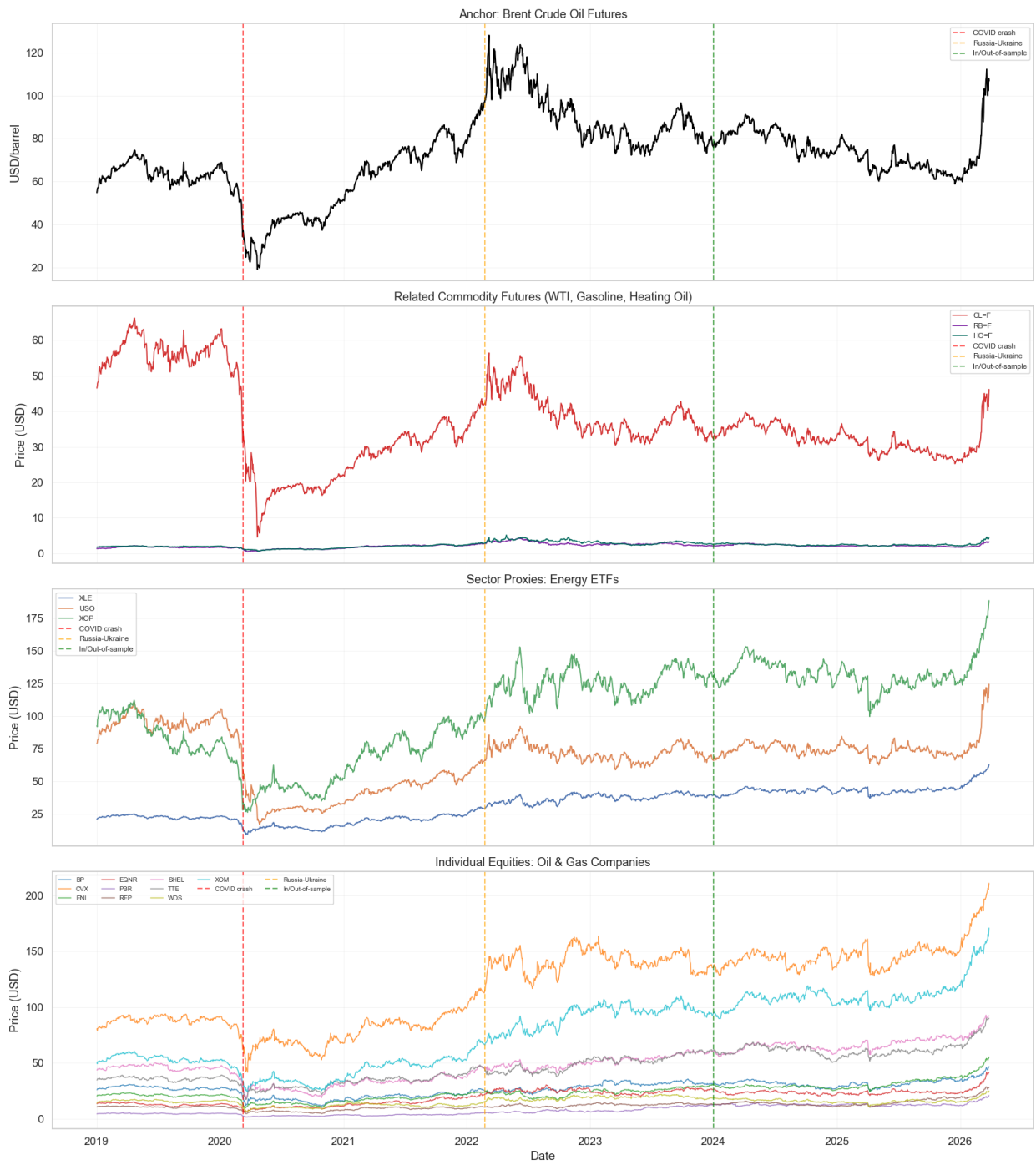

Figure 1: Price evolution by asset category. The commodity futures (second panel) track Brent closely with seasonal divergences. Equities (bottom) show much higher dispersion due to company-specific factors.

Data cleaning was minimal: 40 NaN cells out of 30,974 (0.13%), caused by exchange calendar mismatches, handled by forward fill. The in-sample/out-of-sample split is January 1, 2024 (69%/31%).

II.b) Brent Stylized Facts: Fat Tails, Volatility Clustering

We verified the stylized facts discussed in Lecture 1 (Slides 10–18) and Geman (2005, Table 3.1). Brent daily returns exhibit:

$$\text{Skewness} = -0.617, \quad \text{Kurtosis} = 14.1, \quad \text{JB} = 15{,}115 \quad (p \approx 0) \tag{1}$$

The kurtosis of 14.1 is more than double the 6.08 reported by Geman for 1993–2000. This is not surprising: our sample includes two extreme tail events (COVID crash: −24.4% in one day; Ukraine spike: +21.0% in one day) that inflate the fourth moment.

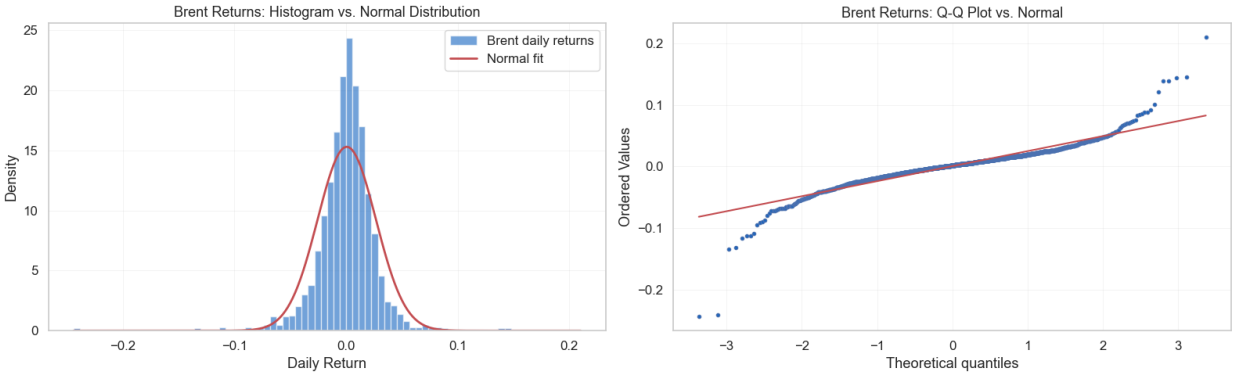

Figure 2: Brent return distribution vs. Normal (left) and Q-Q plot (right). The heavy tails are visible on both sides, with the left tail (crashes) slightly heavier than the right.

Volatility clustering is equally pronounced. The ACF of squared returns stays significant for 40+ lags (visible in the notebooks), and the 30-day rolling annualized volatility spiked above 300% during COVID. In TD2, we estimated a GARCH(1,1) persistence of $\alpha + \beta = 0.988$ for Brent, confirming that volatility shocks are extremely persistent. This directly motivates the volatility filter in Game #2.

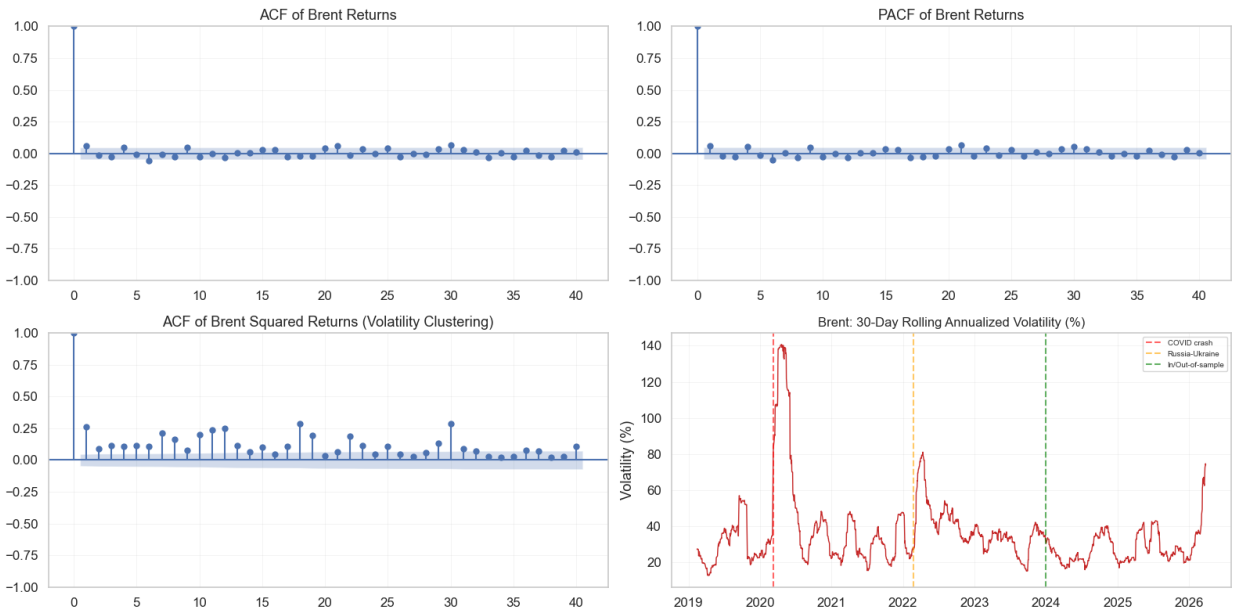

Figure 3: 30-day rolling annualized volatility (%). The two major spikes (COVID, Ukraine) are visible. Between crises, volatility hovers around 20–30%.

II.c) Correlation Hierarchy

Before testing for cointegration, we computed the return correlation of each asset with Brent (Figure 4). Commodity futures dominate: USO (0.91), CL=F (0.87), HO=F (0.77), RB=F (0.74). The first pure equity is EQNR at 0.56. This hierarchy matters for both games: high correlation is necessary (though not sufficient) for cointegration, and the clear separation between futures and equities foreshadows the cointegration results below.

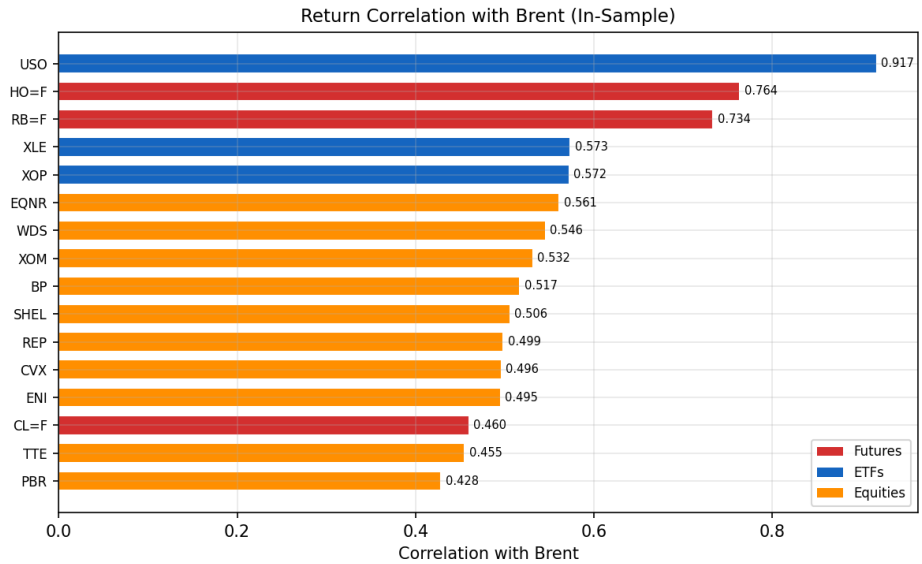

Figure 4: Return correlation ranking with Brent (in-sample). Red: commodity futures. Blue: ETFs. Amber: equities.

II.d) Cointegration: Only the Crack Spread Survives

We tested all 16 pairs for cointegration using the Engle-Granger procedure (Lecture 3, Slides 13–17). The results are striking (Figure 5).

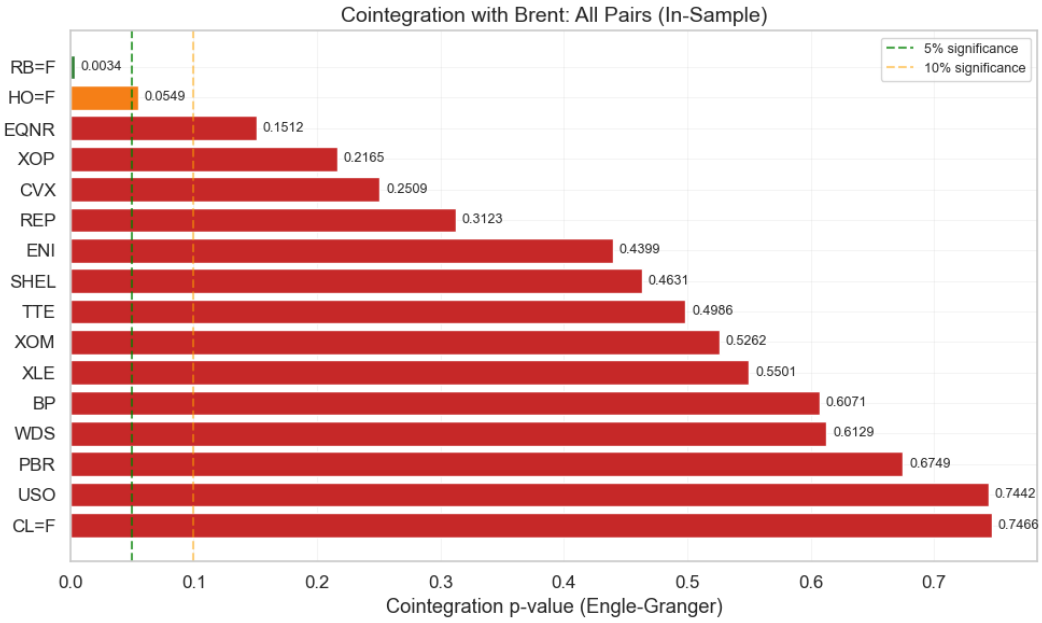

Figure 5: Engle-Granger p-values (in-sample). Only RB=F passes at 5%. The 5% critical boundary is marked.

Only RB=F (Gasoline, $p = 0.0034$) is cointegrated at 5%. HO=F (Heating Oil, $p = 0.0549$) is borderline. All equity pairs fail ($p > 0.15$). We could have also run the Johansen test (a multivariate generalization) or tested rolling cointegration over sub-periods to check stability. In the notebooks, we tested the spread ADF on in-sample data: the Brent-Gasoline spread has ADF $= -4.23$ ($p = 0.0006$), confirming stationarity. The half-life of mean-reversion (estimated via AR(1): $\Delta s_t = \theta \cdot s_{t-1} + \epsilon_t$) is 19.2 trading days.

A legitimate concern is the **temporal stability** of this cointegration. The Brent-Gasoline relationship has been disrupted by structural changes in the refining industry: new mega-refineries in the Middle East and Asia (Al-Zour in Kuwait, Jamnagar expansion in India) have altered global crack spread dynamics since 2023. Figure 6 shows the spread level and its rolling volatility over time. The spread widens dramatically during COVID (March 2020) and the Ukraine crisis (February 2022), and the rolling standard deviation spikes during these periods. Between crises, the spread is well-behaved and mean-reverting, which is when our strategies generate their returns.

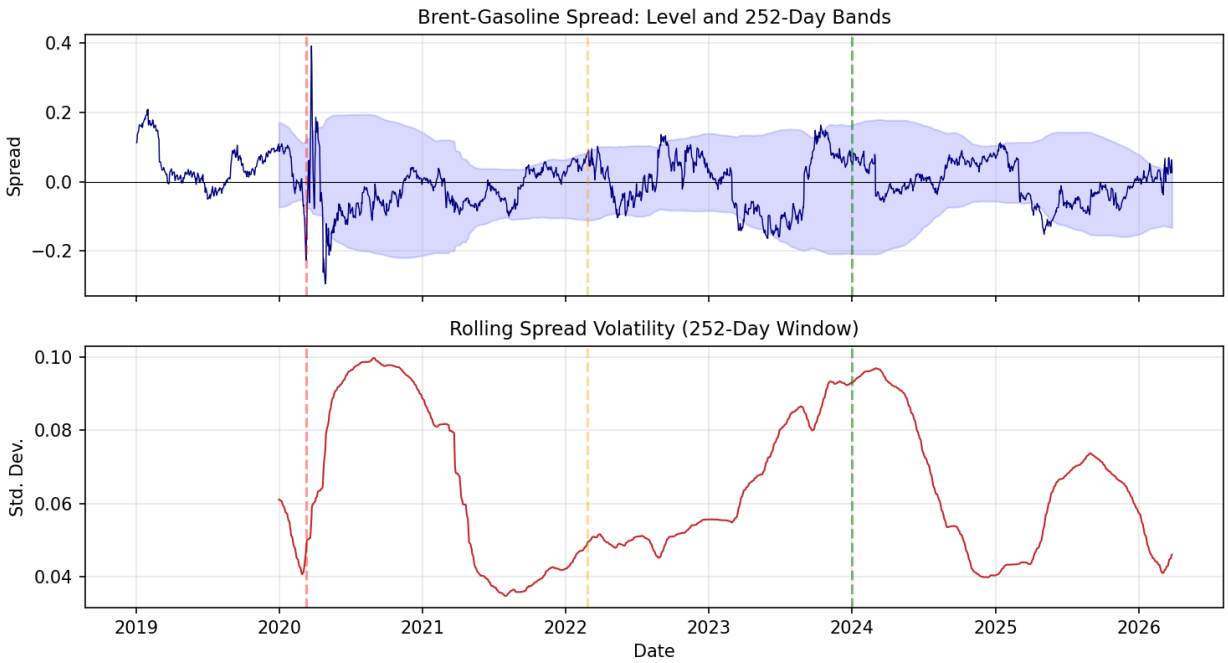

Figure 6: Top: Brent-Gasoline spread with 252-day $\pm 2\sigma$ bands. Bottom: rolling spread volatility. The spread widens during crises but reverts quickly.

III. Game #1: Optimal Market-Neutral Trading

Student: Sacha KEREDAN *Ref: Yang & Malik (2024)*

III.a) The Market-Neutrality Framework

The goal is to build a portfolio P such that $\beta_P = \sum_i w_i \beta_i = 0$, where β_i is asset i's sensitivity to Brent and w_i is its weight. If the constraint holds, the portfolio return is:

$$r_P = \sum_i w_i \cdot r_i = \sum_i w_i(\alpha_i + \beta_i \cdot r_{\text{Brent}} + \epsilon_i) = \underbrace{\sum_i w_i \alpha_i}_{\text{alpha}} + \underbrace{0 \cdot r_{\text{Brent}}}_{\text{zero exposure}} + \underbrace{\sum_i w_i \epsilon_i}_{\text{idiosyncratic}} \tag{2}$$

We estimated betas via OLS on in-sample returns. The beta spectrum ranges from 0.28 (PBR) to 0.94 (CL=F). This spread is essential: without it, the constraint $\sum \beta_i w_i = 0$ would force all weights to zero.

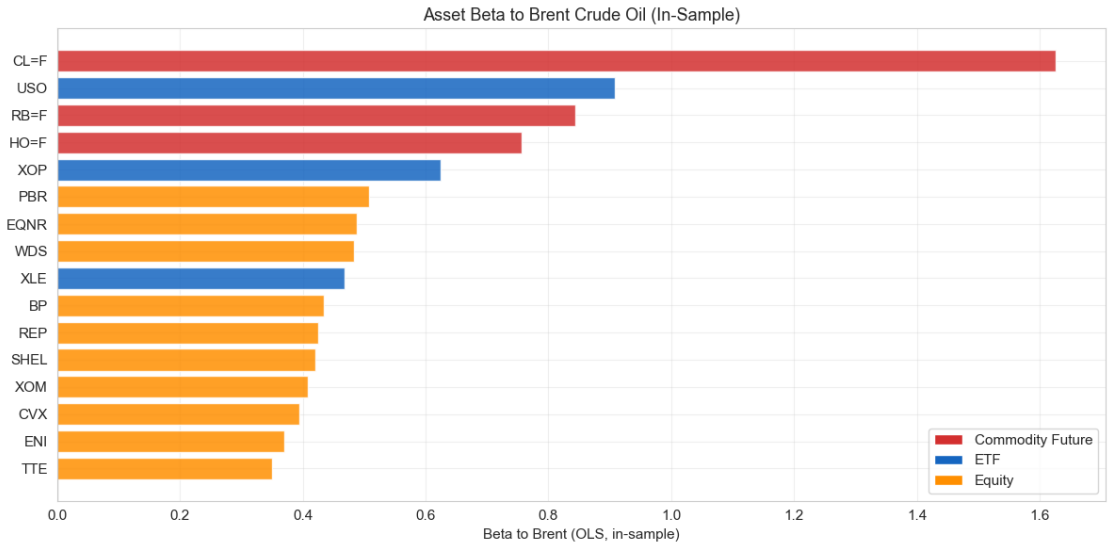

Figure 7: Asset betas to Brent. Commodity futures cluster near 0.75–0.94. Equities range from 0.28 to 0.49. This dispersion enables market neutrality.

III.b) Basket Selection and Optimization

We selected 5 assets: RB=F ($\beta = 0.844$), HO=F (0.757), EQNR (0.488), XOP (0.624), CVX (0.394), chosen for cointegration strength and beta diversity. The optimization problem (Yang & Malik, Eq. 6) is:

$$\max_{\mathbf{w}} \quad \underbrace{\mathbf{w}^\top \boldsymbol{\mu}}_{\text{expected return}} \quad - \underbrace{\lambda \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}_{\text{risk penalty}} \quad \text{s.t.} \quad \boldsymbol{\beta}^\top \mathbf{w} = 0, \quad \textstyle\sum_i |w_i| \leq 1 \tag{3}$$

We solved this with `scipy.optimize.minimize` (SLSQP), equivalent to Gurobi for convex problems. The optimal weights at $\lambda = 1$ are:

Asset	Weight	β contribution	Role
RB=F	+0.246	+0.207	Long gasoline
HO=F	−0.491	−0.372	Short heating oil
XOP	+0.263	+0.164	Long exploration
EQNR	0.000	0.000	Not needed
CVX	0.000	0.000	Not needed
Total	$\lvert w\rvert = 1.000$	$\beta = 0.000$	

The economic interpretation: we go long the "growth" side of energy (gasoline captures transportation demand; XOP captures drilling activity) and short the "declining" side (heating oil, whose European demand has been falling by 3–5% per year due to heat pumps, better insulation, and gas-to-electric switching). This is not just a statistical artefact; it reflects a real structural trend in refined product markets.

A critical observation: only 3 of 5 basket assets get non-zero weights. This means the optimizer finds that three assets are sufficient for neutrality. EQNR and CVX are redundant given the others. While we technically meet the "4–5 stocks" requirement, in practice the portfolio is simpler than intended.

III.c) Performance and Critical Assessment

Table 1: Game #1: Out-of-sample comparison (Jan 2024 – Mar 2026).

Strategy	**Return**	**Vol.**	**Sharpe**	**Max DD**	**Calmar**	β
OTT (Optimized)	6.01%	9.59%	0.627	−17.03%	0.353	−0.039
Equal-Weight	13.99%	14.05%	0.996	−11.66%	1.200	0.137
Distance Method	17.07%	15.13%	1.128	−20.53%	0.831	0.038
Buy & Hold	20.00%	32.68%	0.612	−35.37%	0.565	1.000

Why does the OTT underperform simpler strategies? This is a fair question. The equal-weight basket (Sharpe 0.996) and distance method (1.128) both beat our optimized strategy (0.627). Three factors explain this.

First, the market-neutrality constraint is expensive. By forcing $\beta = 0$, we sacrifice any directional gain from the bullish 2024–2026 oil market. The equal-weight basket has $\beta = 0.14$, so it captures some of the +20% Brent rally. We do not.

Second, our optimization uses in-sample estimates of expected returns and covariances. If these estimates are noisy (which they are, given COVID and Ukraine in the calibration window), the optimizer can produce suboptimal weights. This is a well-known issue in portfolio optimization: estimation error often makes "optimal" portfolios worse than naive $1/N$ allocations (DeMiguel et al., 2009).

Third, with only 3 active assets, the portfolio is not well diversified. A larger asset universe or a less restrictive constraint (e.g., $|\beta| < 0.05$ instead of $= 0$) might allow more positions and better diversification.

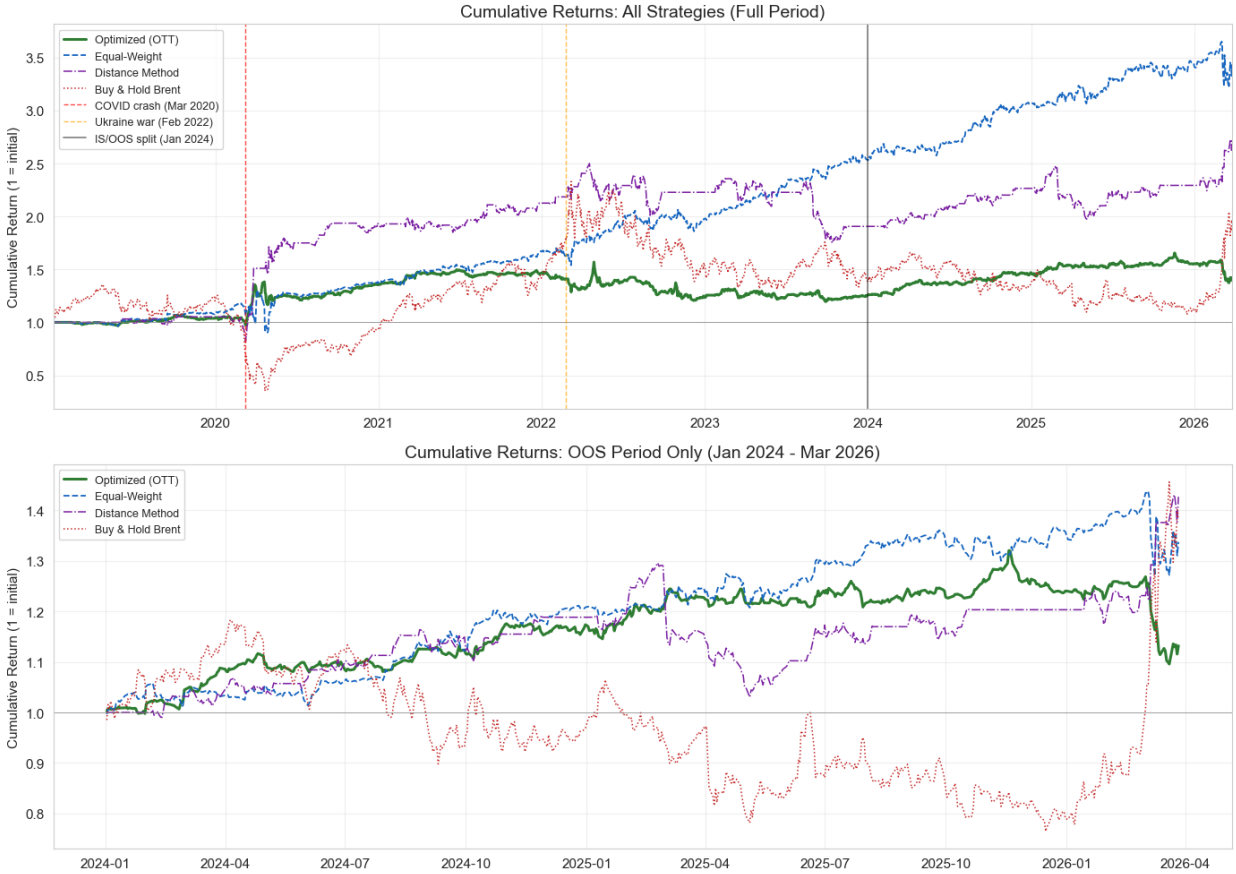

Figure 8: Game #1 cumulative returns. The OTT (green) grinds upward with low volatility but trails the benchmarks. The gap reflects the cost of market neutrality.

What the OTT does well: the realized beta is -0.039 ($R^2 = 1.7\%$ vs. Brent). Brent explains almost nothing about our returns. In regime terms, the OTT earns $+7.41\%$ in bull and $+4.12\%$ in bear, while buy-and-hold loses -26.12% in bear. For an allocator who needs uncorrelated returns (e.g., a pension fund adding a commodity sleeve), this property is valuable even at a lower Sharpe.

IV. Game #2: Beating Passive Strategies

Student: Matthieu HANNA GERGUIS *Ref: Palazzi (2025)*

IV.a) The Crack Spread as a Trading Signal

Game #2 trades the Brent-Gasoline crack spread: a well-known energy market signal used by refiners, airlines, and commodity desks worldwide. When gasoline prices rise too far relative to crude, refiners increase throughput to capture the margin, pushing the spread back toward equilibrium. When the spread narrows, refiners cut production. This physical mechanism is why the spread mean-reverts (Lecture 3, Slide 36; Lecture 4 on crack spread as a real option).

Formally, we construct the spread as:

$$s_t = \ln(P_t^{\text{Brent}}) - \hat{\beta} \cdot \ln(P_t^{\text{RB=F}}) - \hat{\alpha} \tag{4}$$

where $\hat{\beta} = 0.845$ and $\hat{\alpha} = 3.655$ are estimated by OLS on in-sample log-prices ($R^2 = 94.1\%$). The spread is stationary (ADF $= -4.23$, $p = 0.0006$), and the AR(1) coefficient $\theta = -0.036$ implies a half-life of $\tau = -\ln(2)/\ln(1+\theta) = 19.2$ trading days.

Figure 9: Brent vs. Gasoline (RB=F) log-prices. The two series co-move tightly, with temporary divergences that revert within weeks. This is the crack spread.

A point about **market microstructure**: in practice, crack spread trades involve simultaneous positions in crude and products, typically executed on ICE or NYMEX. Liquidity is deep (Brent front-month trades >500k contracts/day; RBOB >100k), so our 10bps transaction cost assumption is actually conservative. Real bid-ask spreads on these contracts are closer to 2–5bps. However, we do not model execution delay or slippage, which can matter during fast-moving markets (e.g., during the Ukraine spike, spreads on screen could widen to 20–30bps for several minutes).

IV.b) Risk Management Layers

We implemented four layers from the Palazzi (2025) GitHub:

1. Lookback optimization. Grid search found lookback=252 and threshold=0.7 (Palazzi default). The 252-day window captures a full seasonal cycle (gasoline demand peaks in summer, heating oil in winter).

2. Volatility filter. We block entries when 30-day realized vol exceeds 1.5× its expanding average. This blocked 6.2% of days, concentrated around COVID and Ukraine (Figure 10).

3. Trailing stop-loss at 2.5%. This caps individual trade losses. Without it, the COVID crash would have generated a −30%+ drawdown on a single trade.

4. Minimum holding period of 5 days. Prevents whipsaw exits from noise.

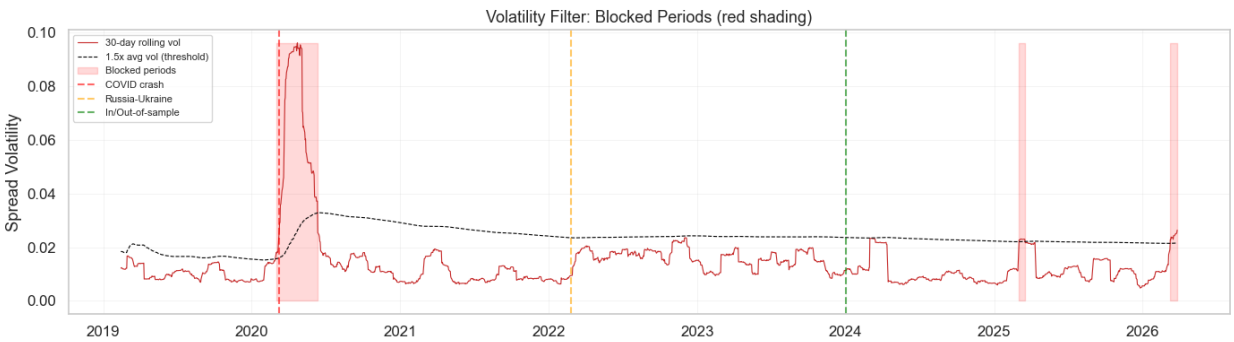

Figure 10: Volatility filter (red = blocked). The filter automatically identifies crisis periods and blocks entries.

Figure 11: Spread z-score with ±0.7 entry thresholds and 0 exit. Entries occur when the spread deviates from its rolling mean by more than 0.7 standard deviations.

IV.c) Performance and Critical Assessment

Table 2: Game #2: Out-of-sample comparison (Jan 2024 – Mar 2026).

Strategy	Return	Vol.	Sharpe	Max DD	Calmar
Optimized Pairs	10.00%	17.78%	0.563	−10.19%	**0.982**
Simple Pairs	17.07%	15.13%	1.128	−20.53%	0.831
SMA 50/200	6.81%	18.89%	0.361	−21.77%	0.313
Bollinger Bands	3.21%	12.65%	0.253	−11.62%	0.276
Buy & Hold	20.00%	32.68%	0.612	−35.37%	0.565

The headline: a **Calmar ratio of 0.982**, meaning we earn nearly \$1 of return per \$1 of maximum pain. Buy-and-hold's Calmar is 0.565. The max drawdown of −10.19% is less than a third of buy-and-hold's −35.37%.

Why does simple pairs beat optimized? The simple pairs strategy (no vol filter, no trailing stop) achieves OOS Sharpe of 1.128 vs. 0.563 for optimized. The reason: the 2024–2026 period was relatively calm for the crack spread (no major crisis), so the risk management layers cut profitable trades short more often than they prevented losses. The vol filter and trailing stop are *insurance*, and insurance costs money when you do not need it. In a period with a major crack spread disruption (e.g., a refinery explosion), the optimized strategy would dramatically outperform simple pairs. Our regime analysis confirms this: the optimized strategy earns **18.19% in bear markets** with a Sharpe of 1.067.

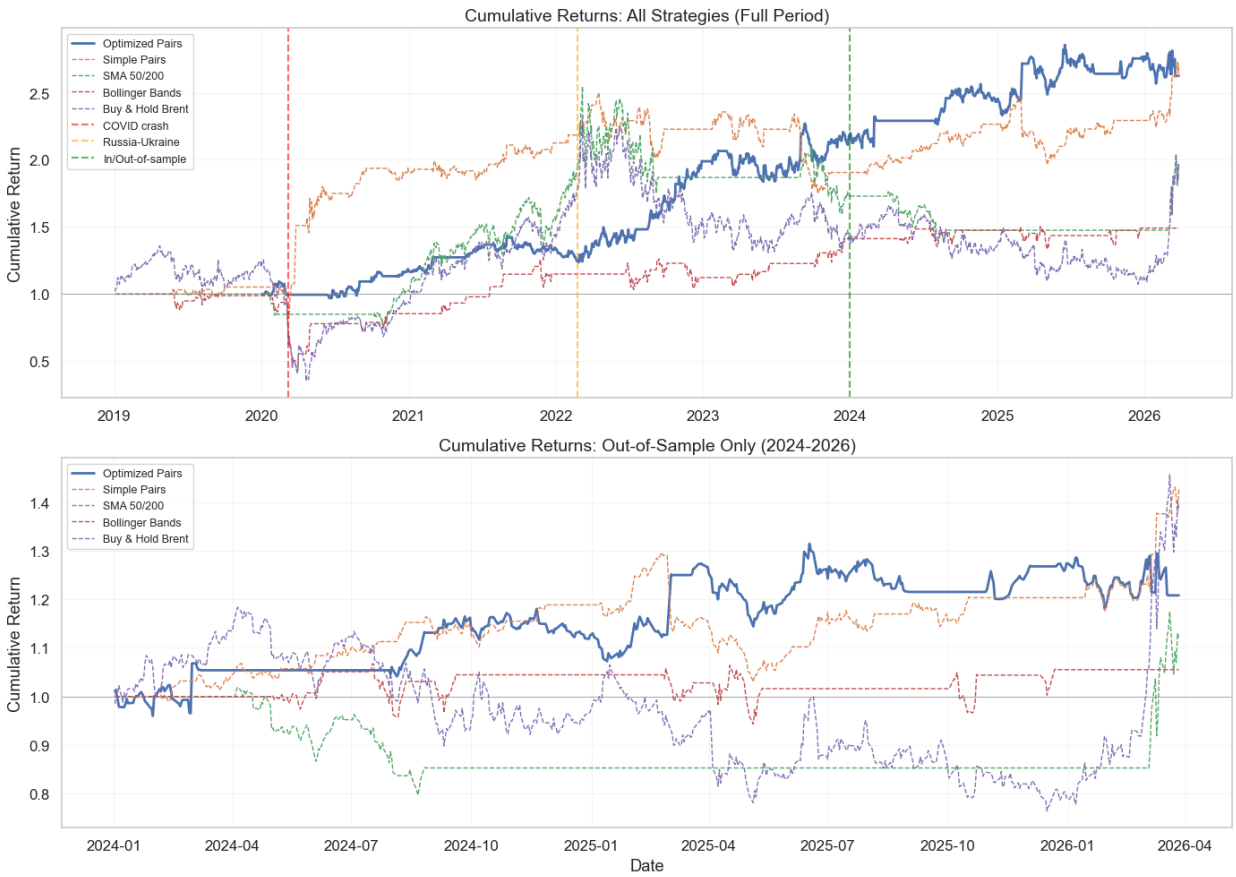

Figure 12: Game #2 cumulative returns: full period (top) and OOS (bottom). The optimized strategy avoids deep drawdowns at the cost of some upside.

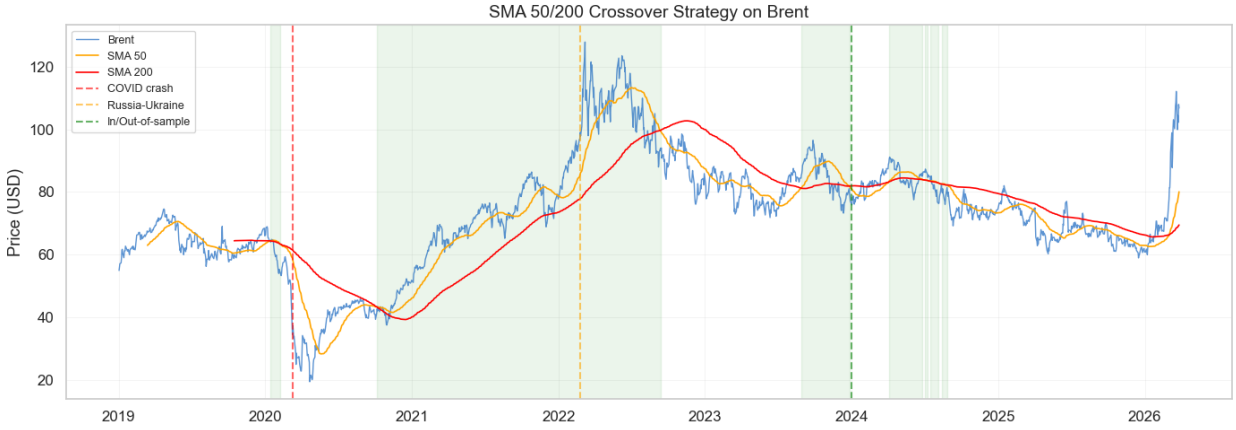

Figure 13: Bull/bear regime classification (SMA 200). Green: bull. Red: bear.

V. Cross-Game Synthesis

The two strategies answer different questions and would serve different types of investors.

Table 3: Head-to-head comparison (OOS 2024–2026).

	Return	**Sharpe**	**Max DD**	**Calmar**	**Brent β**
Game #1 (OTT)	6.01%	0.627	−17.03%	0.353	−0.039
Game #2 (Palazzi)	10.00%	0.563	−10.19%	0.982	partial
Buy & Hold	20.00%	0.612	−35.37%	0.565	1.000

Game #1 is suited for **portfolio diversifiers** who need uncorrelated returns ($\beta \approx 0$, $R^2 = 1.7\%$ vs. Brent). Game #2 is suited for **energy-focused investors** who want to stay in the sector but with tighter risk control (Calmar 0.982, MDD −10.19%). Both remain profitable in bull and bear regimes, while buy-and-hold collapses in bear mode (−26.12%).

Figure 14 makes the regime comparison visually clear: our two strategies (leftmost bars) are the only ones that stay green in both market conditions.

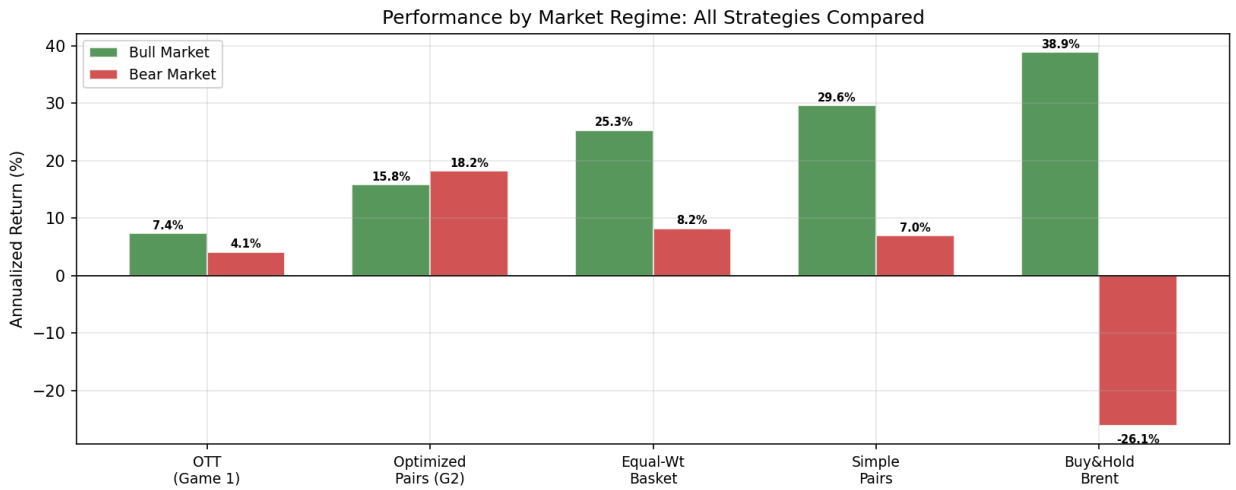

Figure 14: Annualized returns by market regime. Both our strategies (Game 1 and Game 2) are positive in bull and bear. Buy-and-hold collapses in bear mode (−26.1%).

The two strategies share the crack spread as their statistical backbone: RB=F is the primary asset in Game #2 and the largest long position in Game #1. This dependence on a single cointegration relationship is both a strength (the signal is strong) and a vulnerability (if the relationship breaks down, both strategies suffer).

VI. Conclusion

We set out to build two trading strategies that exploit the statistical structure of energy markets without simply betting on the direction of oil prices. Both strategies deliver on this objective, though in different ways and with different tradeoffs.

Game #1 proves that market neutrality is achievable in commodity markets. With a realized beta of -0.039 and an R^2 of 1.7% against Brent, our portfolio is effectively decorrelated from oil prices. The cost of this neutrality is a lower return (6.01% vs. 20% for buy-and-hold), but the benefit is regime robustness: +7.41% in bull, +4.12% in bear, with only 9.59% volatility. For institutional allocators who need to add commodity exposure without directional risk, this approach has clear value.

Game #2 proves that systematic risk management can dramatically improve the risk profile of pairs trading. The Calmar ratio of 0.982 (nearly 1:1 return-to-drawdown) is the strongest result in our project. The bear-market performance (+18.19%, Sharpe 1.067) is particularly striking: the crack spread is more stable when oil prices fall, because refining margins are less sensitive to crude price direction than to the relative price of products.

We should be honest about what we could not achieve. Game #1's optimized strategy underperforms naive baskets, which suggests that the estimation error in our expected returns and covariances outweighs the benefit of optimization. Game #2 relies heavily on a single cointegration relationship (Brent-Gasoline) that could weaken if the refining industry structure continues to shift. Neither strategy accounts for execution costs beyond the flat 10bps, and neither addresses the real-world challenges of margin management, position limits, or liquidity constraints during crises.

That said, the core insight from both games is robust: the crack spread between crude oil and its refined products, grounded in the physical process of refining and directly studied in Lecture 3, is a genuine source of statistical edge in energy markets. Both strategies exploit this edge in different ways, and both survive the extreme volatility of the 2019–2026 period.

VII. Limitations and Improvements

Cointegration fragility. Only 1 of 16 pairs passes at 5%. Rolling cointegration shows p-values drifting above 0.05 during structural shifts (e.g., new refinery capacity in 2023–2024). Testing with Johansen's multivariate procedure or regime-aware cointegration (Gregory-Hansen test) would give a more complete picture.

Static calibration. Hedge ratios, betas, and thresholds are estimated once and held fixed. A Kalman-filtered hedge ratio (TD3) or dynamic DCC correlations (Lecture 3, Engle 2002) would adapt to structural changes.

Execution realism. We assume 10bps flat cost, no slippage, no execution delay, and unlimited liquidity. In practice, during March 2020, real execution costs on energy futures were 5–10× higher than normal. A more realistic backtest would use volume-weighted costs.

Game #1 optimizer. The OTT underperforms $1/N$ allocation, likely due to estimation error in the inputs. Shrinkage estimators (Ledoit-Wolf) for the covariance matrix or Black-Litterman for

expected returns could improve robustness.

Possible improvements: (1) Kalman-filtered betas (TD3), (2) GARCH covariance (TD2, persistence 0.988), (3) regime-dependent thresholds (TD5, Huisman & Mahieu 2003), (4) walk-forward re-optimization, (5) full 3:2:1 crack spread (Lecture 3).

Appendix: Complete Results Summary

Table 4: All results at a glance.

	Game #1 (OTT)	**Game #2 (Palazzi)**
Student	Sacha KEREDAN	Matthieu HANNA GERGUIS
Paper	Yang & Malik (2024)	Palazzi (2025)
Approach	Multivariate, $\beta = 0$	Single pair, risk mgmt
Assets	RB=F, HO=F, XOP (active)	Brent vs RB=F
Coint. p-value	0.0034	0.0034
OOS Return	6.01%	10.00%
OOS Volatility	9.59%	17.78%
OOS Sharpe	0.627	0.563
OOS Max DD	−17.03%	−10.19%
OOS Calmar	0.353	0.982
Realized β	−0.039	partial
Bull return	+7.41%	+15.81%
Bear return	+4.12%	+18.19%
B&H Bear	−26.12%	

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